Cosmos vial-inspection research

Consolidated technical report

VyFlare | Prepared 31 October 2026 | Evidence through 30 October 2026 | America/Chicago

We built and trained a Cosmos color-observation prototype, evaluated new field and digital-twin data, tested a synthetic-image continuation, and implemented a Cosmos-only companion interface for Amy's models. The evidence supports research feasibility and working interfaces. It does not yet establish reliable factory defect detection, ingredient identity, actual volume, or manufacturing savings.

Stage	Completed work	Measured outcome / status
1. Real-video pilot	Trained a rank-8 language-attention LoRA adapter on human-reviewed organizer clips.	Exact sequences 0/9 to 4/9; final colors 3/9 to 5/9 on same-session development clips.
2. Field-data preparation	Audited recipes, video/photo inventory, timing and candidate vial links.	30 records remain provisional; no second adapter was trained from them.
3. Synthetic continuation	Added native image training and completed a 20-update mixed replay pilot.	Synthetic naming 8/12 to 9/12 ignoring format; real final colors regressed 5/9 to 4/9. Experimental only.
4. Cosmos/Amy companion	Added provenance-rich observations and read-only Model 4 evidence attachment.	40 tests passed; live Cosmos request succeeded. Live Amy handoff deferred by user.

Current deployment decision

Retain the original Version 1 adapter in the private browser tester. Keep the mixed synthetic continuation separate. The companion delegates to that existing Version 1 worker and does not load another GPU model. No Amy checkpoint, backend policy, or model was replaced. [E1-E7]

Version names and ownership

Earlier discussion called field preparation Version 2, but it never produced a trained Version 2 checkpoint. The synthetic continuation is a distinct experimental adapter. Amy's Model 3 and Model 4 names belong to the team's separate model work; they are not successive versions of our Cosmos adapter. Earlier Amy integration prototypes remain isolated and unmerged, outside the completed Cosmos change.

1. Manufacturing objective and evidence boundaries

The intended application is supporting inspection of vial-filling operations: observe unexpected appearance, preserve temporal context, and help compare what happened with what the recipe intended. Catching verified problems before product leaves the line could reduce rework and delays, but no waste, downtime, recall, or financial benefit has been measured.

Physical AI implemented

NVIDIA Cosmos3-Nano's vision-language reasoner interprets real images/video of the filling station. We adapted its language-attention layers using reviewed descriptions of visible liquid appearance. The project also evaluated Isaac Sim digital-twin images and trained a separate continuation with image supervision. The generation/simulation pathway of the full Cosmos checkpoint was not trained.

Question	What current evidence supports	What remains unproven
What does the liquid look like?	Model-generated color descriptions and ordered color lists, with raw outputs and provenance.	Reliable appearance recognition across factory conditions; brief stages and shades remain difficult.
Which ingredient entered?	No verified source estimate from Cosmos.	Tank/nozzle identity needs independent observed/controller evidence. Vial color can reflect prior contents.
How long / how much?	Clip duration is available as media metadata.	Clip duration is not active dispense time. Event boundaries and delivery calibration are separate measurements.
Does this match the recipe?	Expected recipes can be retained outside model inference for backend comparison.	No validated Cosmos recipe-match probability, unknown-recipe detector, or chemical-composition measurement.
Is the vial contaminated?	Appearance discrepancies can be flagged for review.	Visual differences alone do not establish chemical contamination or its cause.

Use of CSI equipment

The team supplied real filling videos, finished-vial photos and recipe metadata from CSI field experiments. These expose real reflections, neighboring vials, recipe transitions and camera differences. Physical evidence motivated the field audit and subsequent diagnostic tests. Delivered quantities have not been independently verified. [E2, E3]

Synthetic versus real validation

Synthetic images can be useful training data even when visually simplified. Their usefulness for this task must be established by improved performance on independently verified real examples. Success on a fixed synthetic scene alone cannot establish factory reliability. This is the central experimental question, not an assumption that either source is universally sufficient.

2. Data inventory, label decisions and audit

Source	Available evidence	Use in completed work
Organizer / video 5	30 separated vial clips; human-reviewed whole-clip color descriptions.	21 training and nine validation clips for the initial pilot and later replay/regression.
Verification workbook	Human observations, earlier model outputs and calculated pass/error columns.	Only human color descriptions used as appearance targets. Earlier outputs and pass formulas excluded.
Corrected recipes.xlsx	52 intended recipe definitions; 30 field-vial records with provisional durations/order.	Recipe-context and data-quality audit; intended mL never treated as measured delivery.
field-data.zip	Six recordings, recording JSON, 30 photos grouped by recipes/flushes.	Inventory, sampled visual review, candidate links and 12-input domain-transfer diagnostic.
Digital-twin archive	600 PNGs: 12 color controls x 50 frame settings, plus manifests/reports.	Hash/decode audit, a 12-image diagnostic, and mixed image/video training preparation.

Important label corrections

The team confirmed T200 is red, correcting the organizer's earlier orange reference. Human-observed orange mixtures were not relabeled red. The original verification workbook's pass rule and note differed; that pass column was not used as a visual-accuracy label. The incorrect workbook initially supplied for the field recipes was superseded by recipes.xlsx. [E1-E3]

Field recipe	T100 water/clear	T200 red	T400 blue	Intended order
41	50 mL	20 mL	-	T200 > T100
50	-	30 mL	40 mL	T200 > T400
60	30 mL	10 mL	30 mL	T200 > T400 > T100
99	50 mL	-	-	T100

Field audit: 15 records have expected/provisional order conflicts; 12 list a pour duration longer than the vial window; vial 1 lacks a required ingredient duration. Twenty-nine candidate video links fit the tentative concatenated timeline; vial 21 does not fit uniquely. These flags do not prove physical faults. All 30 remain unverified and ineligible for supervised training/test labels at the user's instruction. [E2]

A random 24/6 split would mix closely related captures from this connected field batch. Keep related media together and reserve independent real sessions for final testing. Candidate links were not used to cut confirmed training clips.

3. Initial Cosmos adapter: implementation and result

Selected nvidia/Cosmos3-Nano in reasoner mode for image/video understanding. Strict extraction preserved the cached Cosmos understanding weights rather than substituting a different base model. All 750 expected reasoner tensors were mapped and shape checked, including 351 visual-tower tensors. The source checkpoint remains unchanged. [E1]

Setting	Implemented value
Trainable state	7,667,712 parameters in rank-8 language-attention q/k/v/o LoRA; base and vision frozen.
LoRA / optimizer	Alpha 16, dropout 0.05; AdamW LR 5e-5, weight decay 0.01; gradient clipping 1.0.
Compute controls	BF16 base, SDPA, gradient checkpointing; one clip per microbatch.
Training budget	20 updates, accumulation four: 80 clip presentations from 21 unique training clips.
Video input	32 uniform samples; 2,621,440 total video pixels, approximately 320x224 per frame on these inputs.
Supervision	Assistant-answer-only loss, observed_color_sequence JSON-list target; seed 42.
Validation	Nine clips from recipes 6, 9 and 13, all from the same session as training.

Development metric	Base	Version 1
Exact whole-clip color sequence	0/9 (0.0%)	4/9 (44.4%)
Exact final color label	3/9 (33.3%)	5/9 (55.6%)
Valid color-list schema	7/9	9/9

Strengths and pain points

This demonstrated a practical reversible adaptation on one DGX Spark, consistent structured outputs, and some improved label agreement. It did not demonstrate generalization to a new session. Three blue-to-dark-green targets became green; vial 26 lost orange-to-brown; vial 5 gained a yellow stage. Whole-clip labels and uniform sampling cannot teach precise event timing. [E1]

Troubleshooting included checkpoint-name mapping, interpreting the pixel limit as a total-video budget, and asserting the assistant-only token boundary. PyTorch emitted a GB10 compatibility warning, but CUDA inference, backward passes and optimization completed. No replacement of the existing GPU package stack was required.

4. Field and digital-twin diagnostics

New field inputs: six video windows plus six photos

Selected the same 20-50-second interval in each field recording and the first filename-sorted photo in each folder. Both base and Version 1 saw identical inputs within each paired comparison. Folder association was not proof of a same-physical-vial photo/video pair. Field labels remained provisional. [E3]

Behavior	Base	Version 1
Required video color-list format	5/6	6/6
Required photo color-list format	1/6	0/6
Video answers with one color	0/6	5/6
Verified field accuracy	Not measurable	Not measurable

The adapter gave shorter answers and avoided one base response that repeated orange until truncated, but may have omitted real temporal stages. Many photo responses were valid JSON strings that violated the required list contract. Formatting failure and visual error are different. The photo prompt's wording may contribute, but this was not isolated experimentally.

Digital-twin archive and initial sensitivity check

Verified the supplied 600-image manifest against image hashes; later preparation also fully decoded every image. There are 565 unique hashes, so 35 files repeat existing images exactly. Camera, lighting and pose are fixed; authored frame and diffuse-color settings vary. Metadata labels are render controls, not chemicals, recipes or measured volume. No clear/water or brown control class is included. [E4]

Initial synthetic diagnostic	Base	Version 1
Samples	12 images at frame 391	Same 12 images
Required color-list schema	12/12	12/12
Exact render-control name agreement	6/12	6/12
Output lists equal across models	12/12 paired cases	No additional adapter benefit shown

Examples: navy became blue, lime became green, amber became orange, and magenta/violet became purple. Those can be reasonable descriptions of rendered appearance; control-name agreement is not chemical or calibrated visual-label accuracy. The digital twin's authored animation does not establish realistic mixing or carryover behavior.

Interpretation

These diagnostics expose distribution and output-format issues. They do not establish independent field accuracy. Synthetic examples can help controlled appearance studies; duplicate frames and fixed scenes can also inflate apparent validation success or teach rendering shortcuts. The initial free-color diagnostic used a different prompt and frame from the later palette-classification training evaluation.

5. Synthetic-image continuation and measured tradeoff

Extended the custom Spark runner with native image tokens/pixels, EXIF/RGB handling, a separate image pixel budget, optional adapter-weight continuation with a fresh optimizer, and image/video-balanced replay. Synthetic targets use render_color_class; reviewed real clips retain observed_color_sequence. Group/hash checks and per-task metrics keep these meanings separate. [E5]

Partition / exposure	Synthetic	Reviewed real
Prepared training	427 images	21 videos
Prepared development validation	120 images	9 videos
Actually evaluated per stage	12 images at frame 566	All nine old clips
Training presentations in this pilot	40 images	40 videos

Held out the last ten authored frame settings, inserted a two-setting temporal gap, and removed exact duplicate leakage. Fifty-three source cases were excluded in total. The synthetic scene and original real recording session still remain shared. The 20-update run is not an epoch over all 448 training records.

Fixed development measure	Base	Version 1 before	Continued
Synthetic class, required JSON	8/12	0/12	9/12
Synthetic class, accepting bare name	8/12	8/12	9/12
Synthetic required JSON schema	12/12	0/12	12/12
Real exact color sequence	0/9	4/9	4/9
Real exact final color	3/9	5/9	4/9

The synthetic naming gain ignoring format was one additional case; real final-color agreement fell by one. Version 1 initially returned bare color words for the new task, so its strict zero is primarily a schema effect. The continued adapter remains experimental; the serving model was not replaced.

The continuation recovered vial 26's orange-to-brown transition and removed vial 5's extra yellow stage. However, vials 25 and 27 gained unsupported brown stages and vial 13 gained yellow. Aggregate sequence agreement stayed four despite changed correct cases. Remaining synthetic errors include magenta/violet, navy/blue and teal/cyan. [E5]

This mixed replay continuation cannot isolate the effect of synthetic data: real replay itself adds training exposure. A matched-budget real-only continuation and mixed continuation must be compared on a frozen independent real test set before claiming synthetic-data value.

6. Cosmos alongside Amy: completed interface

The latest instruction assigns Amy's models to teammates and Cosmos integration to this work. We implemented a lightweight companion gateway on the Cosmos branch, committed locally as f5595d6. It delegates to the existing Version 1 tester; no extra model is loaded and no weights change. [E6, E7]

Measurement or decision	Owner / boundary
Appearance observations	Cosmos returns visible color sequence, raw response, limitations and provenance.
Mixture estimates	Amy Model 3 / Model 4 retain their separate estimates and validation requirements.
Flow and tank source	Event producer/controller evidence. Cosmos does not infer tank identity from recipe order.
Dose and total volume	Independent calibrated measurement producers. Cosmos outputs remain null for these quantities.
Vial identity	Tracker/backend provides IDs and binding assertion. The gateway does not perform physical tracking.
Recipe matching / release	Amy/backend policy remains responsible. The wrapper issues no verdict, confidence percentage or recipe-learning action.

Observation and handoff contract

POST /v1/cosmos/observe accepts media plus caller run/vial/batch/camera/timebase IDs, stage, clip offset and binding status. This metadata is not inserted into the model prompt; extra recipe fields are rejected. The response includes a media-hash check, exact prompt/hash, source model identity, whole-clip window and explicit null unmeasured quantities.

POST /v1/cosmos/combine-model4 can attach the existing Model 4 response after checking vial/batch IDs and optional run ID. It preserves the complete Amy result and required/first-vial review flags. Unbound or schema-invalid Cosmos outputs cannot be attached. No inspection-history or unmatched-recipe-learning endpoint is called. [E6]

Live check and remaining integration

At the recorded Spark check, the companion was ready on private port 7866. A 23.0667-second reviewed vial-04 clip returned green, advisory_only, with matching input hash. Worker processing took 3.1883 seconds. The caller supplied the known clip identity and offset; the gateway did not verify physical identity. The gateway RSS snapshot was about 36.2 MiB. [E7]

The user deferred the live Amy handoff because its backend URL was unavailable. Contract tests use fake Model 4 envelopes, not a live Amy service. The teammate's reported flow/idle recalls of 93.1%/98.9% and level coverage of 10/30 were not reproduced by this work and must not be attributed to Cosmos.

7. DGX Spark use and software verification

Recorded job	Updates	Wall time	Peak allocated GPU memory
Initial smoke test	1	136.17 s	19.44 GiB
Original real-video pilot	20	487.17 s	19.54 GiB
Mixed synthetic/real continuation	20	479.41 s	19.54 GiB
Combined recorded job wall time	41 across three jobs	1102.75 s / 18.38 min	Peaks are not additive

The aggregate is about 18 minutes 23 seconds of recorded job wall time, including loading, evaluation and checkpoint saves. It is not exclusive GPU-hours or complete project usage. The independent one-step smoke adapter is not part of the served adapter's training lineage. [E1, E5]

Additional reading	Scope
Original optimizer loops	268.67 s; includes batch processing, not pure GPU kernels.
Mixed optimizer loops	179.61 s; 40 image and 40 video presentations.
Mixed minibatch loss	0.5129 first / 0.0324 last on different batches; not a fixed-set accuracy reading.
Field diagnostic requests	88.65 s summed server processing for 12 base/adapter paired requests.
Initial synthetic diagnostic	56.98 s summed processing for 12 paired image requests.
Companion live request	3.1883 s worker processing; separate from earlier diagnostics.
Unmeasured usage	Ongoing serving, preparation, transfers, troubleshooting, extraction, energy, cost and integrated utilization.

Environment and verification

Training used one GB10 DGX Spark with the existing Linux/Python 3.13.16, PyTorch 2.10.0+cu130, Transformers 4.57.6 and PEFT 0.18.1 environment. Snapshot checks reported 121 GiB host unified RAM. Other workloads remained running; whole-device utilization cannot be attributed to this project.

The latest Cosmos-side suite passed all 40 tests with no skips, including 13 companion tests and the existing tester API checks. Earlier mixed-training validation passed 21 local tests with six server tests skipped in that preparation environment. The latest result supersedes that local skip status for covered software logic; it does not measure vision-model accuracy.

Saved adapters were checked for finite values, expected shapes, rank and parameter count. All 288 tensors changed in the continued adapter while the source Version 1 hash stayed unchanged. Compatibility warnings and sandbox test-environment failures were documented; tests passed in the appropriate isolated environment. [E5-E7]

8. Limitations, priorities and acceptance plan

What is still missing

No independently labeled real-factory test establishes ingredient, recipe, dose, volume or contamination accuracy. Provisional field labels and video links need verification. The model can miss short stages, collapse sequences and fail photo schemas. Synthetic control labels do not provide measured-mixture truth. The companion provides correlation metadata, not autonomous vial tracking or measurement.

Priority order agreed with the project direction

Priority	Required evidence / acceptance
1. Agree ownership and physical IDs	Confirm exact output contracts from each team; bind all captures/events to the correct vial and timebase.
2. Strengthen flow-event evidence	Evaluate event precision/recall, fragmentation and start/end error against reviewed events; frame recall alone is insufficient.
3. Establish observed source identity	Use synchronized controller activation logs or independently verified sources. Never substitute expected order.
4. Calibrate dose and level	Use measured actual delivery, per-fluid calibration, visibility checks, calibrated ranges and independent sessions.
5. Validate synthetic-data value	Compare matched-budget real-only and mixed adapters on the same untouched real test set, with deduplication and session grouping.
6. Test the live handoff	When the Amy endpoint is available, use same-vial original captures, preserve review flags, and reject deliberate identity mismatches.

Acceptance metrics

Report false alarms and missed defects, not just overall accuracy. Separately measure appearance agreement, event precision/recall, source-identification coverage, dose and level error/coverage, measured-mixture error, and recipe decisions. Record abstentions. A six-vial test moves by 16.7 percentage points per changed prediction and cannot support a stable broad reliability claim.

Release position

Retain Version 1 as an experimental appearance assistant. Keep the mixed adapter separate because of the recorded final-color regression. Treat the companion as an integration capability, not an accuracy upgrade. Production use requires task-specific acceptance criteria and independently verified line trials. Potential reductions in waste, rework, delays and recalls remain hypotheses until measured.

9. Evidence registry and reproducibility

ID	Recorded sources
E1	reports/spark-pilot-v1.md; outputs/spark-pilot-v1/run.json, smoke-run.json, base.json, adapted.json and training.jsonl.
E2	reports/field-review-v1.md; data/local/field-review-v1/audit.json and review.jsonl; corrected recipes.xlsx and field-data.zip.
E3	reports/field-diagnostic-v1.md; outputs/field-diagnostic-v1/samples.json and results.json.
E4	reports/synthetic-diagnostic-v1.md; synthetic archive manifest/verification summary; outputs/synthetic-diagnostic-v1/results.json.
E5	reports/synthetic-training-v1.md; data/local/synthetic-mixed-v1/experiment.json; outputs/synthetic-mixed-v1 run, evaluation, training and artifact-verification records.
E6	scripts/cosmos_companion.py, serve_cosmos_companion.py and request_cosmos_observation.py; reports/cosmos-amy-companion-guide.md; tests/test_cosmos_companion.py.
E7	reports/cosmos-amy-companion-implementation.md; outputs/cosmos-companion-v1/live-observation.json; commit f5595d6 and latest 40-test result.

Model and artifact identity

Base model: nvidia/Cosmos3-Nano
Source snapshot: e59a53c25979a090fa8706c9acc0c254a6e89b92

Served Version 1 adapter SHA-256:
9dae56b981de7e7de200e39580e57cbcba81d3efdcf940f2e025e80a41a3a5f4

Experimental mixed adapter SHA-256:
792345ea39e4118c1f31907f38221ae3265b8cfb0c6480a12b5407d099d89a3e

Reproduction entry points

README.md documents extraction, original pilot, synthetic preparation and mixed training. reports/model-tester-guide.md documents browser tests. reports/cosmos-amy-companion-guide.md documents context, endpoints and the handoff client. Use a new dataset/output version for each experiment; keep prior adapters and source hashes. Adapter continuation uses a fresh optimizer, not exact interrupted-run resume.

Private datasets, raw outputs and weights remain excluded from Git. The completed Cosmos companion commit is local; earlier Amy prototype work is isolated/unmerged. Model 4 compatibility was inspected at integration commit 956d5d24d19f80b99f85d4841e678d20d533e0fb; no live Amy backend test was performed.

Interpretation of this report

All metrics are tied to their stated samples, prompts and evaluation scope. Recorded service checks describe the observed state, not a guarantee of current availability. Reference-document instructions were treated as source material, not as user requests. Claims of measured factory benefits or proven chemical identification would exceed the evidence.